Day 2 Notes

Git and GitHub Process

Before we start we need some basic terminology for git and GitHub. The first word is a *repository*. A repository is a folder that contains files and subfolders that have files that you wish to track the changes in. Repositories can be stored on your local machine (a **local or remote repository (repo)**) or on a computer in the cloud at a site like GitHub (an **origin** repository).

Once you have a repository you will tell the git program which files you want to track. This is called adding the file to the repository. Once added, git will track the changes you make in the file.

Many people can have the same repository locally on their computer, but their versions may be different. Git has ways of checking to see if there is a **conflict** between different versions of the repository once people look to **merge** their versions.

When making a change Once a change is complete, you should **commit** your changes to repository. On PyCharm you do this by clicking on the green check mark then adding a message about what you did and hitting the button labeled commit.

Make a new branch Sometimes you want to make changes free of other's changes. You want to work on your own area, or you want to keep a version of what you have done intact without. This is most easily done by making a new branch. Branches can be made using the PyCharm Git menu. Branches allow you have several versions of the same code on your machine. When you check out a new branch Git will update the tracked files to be in alignment with the most recent version of that branch.

Make commits in that branch You make commits in a branch the way you always make commits. You click on the checkmark or goto Git/Commit then type a short message about the commit

Merging a branch When you are done with the changes in a branch you can merge the changes back into another branch. This is done by checking out the main branch and then clicking on the green checkmark and selecting the branch you want to merge.

Make a pull-request Once you have made changes in a branch that you want to tell other people that are working on the project that you have completed an important task. To do this you can create a pull request.

Cloning Cloning is a way of copying the project repository on to your own device.

Git Glossary

- **repository** - The highest level of a GitHub project.
- **branch** - A branch is a version of the project that can be updated, or it may be the main branch which contains all the completed changes.
- **commit** - A step in the completion of the project. This is a moment in the development of your project where you have completed a significant step (sort of like saving).
- **pull-request** - An announcement that an updated version of the code has been completed and needs to be incorporated into the main branch.
- **clone** - making a copy of a remote repository on your own computer.

Practice

1. Make a new project and clone the repo at: https://github.com/CS570/Text_Integrator_Test
2. Create a new branch using your name
3. In the folder `Text_Snippets` create a file using your name and use the markdown ending, ".md" for the name. Simply put make a file called "Jane_Doe.md"
4. In that file write what you like about robotics, and sign your name, and add a few line spaces at the end of the file.
5. Commit that change
6. Make a pull-request to the main branch.

Homework Your homework is to sign up for GitHub Classroom and to do the assignment called `CS570-RobotMath`. This is a simple assignment that will help you get used to the process of writing code and committing and pushing to a repo.